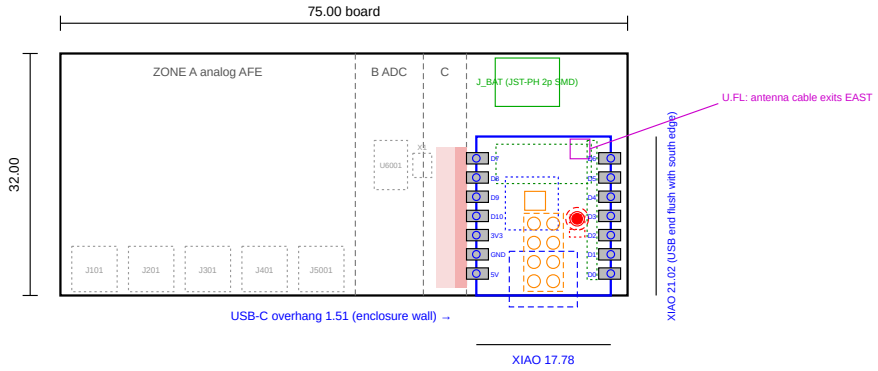
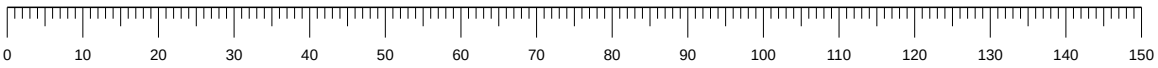
A.R.T.H.U.R EMG Rev B — XIAO ESP32-S3 direct-mount check drawing (RB-00.6)

PRINT AT 100 % / ACTUAL SIZE — NO 'FIT TO PAGE'. This page is 1:1. Verify both rulers before use.



- Rev B hand-solder castellation pad 1.50 × 2.90 (toe 1.30 outside XIAO edge, 1.60 under); 14×, pitch 2.54
- XIAO ESP32-S3 body 17.78 × 21.02, castellations (circles), ESP32 (dotted), USB-C receptacle (dashed)
- BAT+ post: plated hole Ø1.40 drill under XIAO BAT+ pad; top ring Ø1.90 (masked); bottom VBAT pad Ø3.00 (dashed)
- XIAO exposed bottom pads 15–22 + 25: NO F.Cu copper, NO vias, solder mask only (dashed = keep-out)
- Tented-via field (dotted) and D0–D3 F.Cu routing channel east of BAT+ (dotted)
- J_BAT battery connector (approx. envelope, north strip, Zone D)
- No component within 1.5 mm of toe pads; pink-light: parts ≤ 1.2 mm tall only (1.5–4.0 mm)
- Indicative neighbours: electrode connectors, U6001, X1 (final positions in RB-02)

SCALE CHECK 1: this ruler must measure exactly 150.0 mm (±0.5) end to end.



SCALE CHECK 2: the square below must be exactly 50.0 × 50.0 mm; the XIAO must match its 17.78 × 21.02 outline above.

50.0 × 50.0

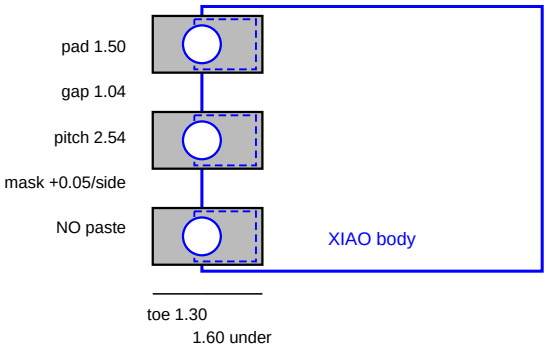
PAPER FIT CHECK (before ordering PCBs):

1. Measure both scale checks. Reject print if error > 0.5 mm.
2. Lay a real (header-less) XIAO ESP32-S3 on the blue outline, USB-C end on the board's south edge line.
3. Each castellation must sit on a grey pad centre (±0.2 mm) and ≥ 1.0 mm of grey toe must remain visible on BOTH sides.
4. XIAO BAT+ pad (bottom) must sit over the red hole.
5. U.FL must fall inside the magenta square.
6. Record results in REVB_XIAO_MECHANICAL_CHECKLIST.md.

Board frame origin = north-west corner; +x east, +y south.

Detail sheet — 5 : 1 (NOT FOR PAPER FIT CHECK)

A. Hand-solder castellation pad (west side shown; east side mirrored)



grey = Rev B pad (HASL, no stencil paste)

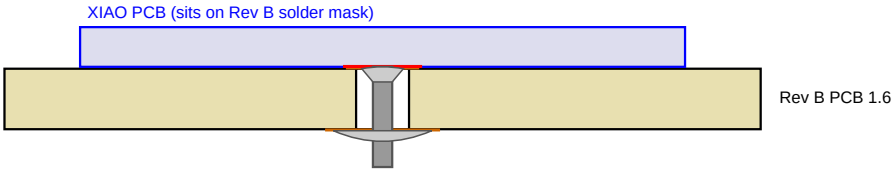
blue = XIAO castellation half-hole + bottom pad

(module pad 1.626 × 1.325, ≈0.2 past edge)

Fillet: wets ≥ 1.0 mm of toe and climbs the

castellation wall ≥ 50 % of XIAO PCB thickness.

B. BAT+ post cross-section (through Rev B PCB, 1.6 mm)



Post: AWG 24 solid tinned Cu (Ø0.51), straight, soldered ⊥ to XIAO BAT+ BEFORE mounting.

Solder cone on XIAO pad: centred, Ø ≤ 1.1 mm, height ≤ 0.5 mm (sinks into the Ø1.3 finished hole).

Hole: plated, drill Ø1.40 (finished ≈ Ø1.30); radial clearance ≈ 0.4 mm = locating pin for the XIAO.

Top ring Ø1.90 is covered by solder mask (tented ring, hole open); 0.45 mm from XIAO GND0 pad edge.

Bottom: Ø3.00 VBAT pad on B.Cu, thermal-relief to VBAT; solder from underside, trim post to ≈ 1 mm.

No bend, no insulation needed: no exposed length exists between XIAO and Rev B (gap ≤ 0.1 mm).